

Tushar Jain

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Summary

Computer Science undergraduate and Backend AI Engineering intern with hands-on experience building REST APIs, retrieval-grounded AI systems, and data-driven dashboards. Comfortable owning the full stack – from API contract design and database modeling in Node.js/Express and Supabase, to responsive React frontends. Seeking remote, high-ownership engineering roles where I can keep building production backend systems.

Experience

Backend AI Engineer – Intern

June 2026 – Present

FlyRank AI

Chicago, Illinois, United States (Remote)

- Built and tested REST API endpoints in Node.js/Express, defining request/response contracts, payload shapes, and error states before implementation.
- Integrated Supabase as the backing data store, writing queries and data models to support endpoint logic and downstream search features.
- Worked on retrieval-oriented backend logic to ground AI-generated responses in real source data rather than unsupported output.
- Shipped a capstone multi-platform publishing system (adapter interface, idempotent retries, rate-limit backoff, encrypted credentials, verified webhooks) – detailed under Projects.

Projects

Social Media Studio | *Next.js, TypeScript, Express, SQLite, Sharp*

- Built a multi-platform social publishing pipeline (Instagram, X) with a durable SQLite-backed job queue and adapter-pattern publisher interface, covered by 15 passing tests across 4 suites.
- Implemented idempotent publish handling via idempotency keys, ensuring retried requests never double-post to a platform.
- Added rate-limit-aware retry logic honoring 429/Retry-After responses with exponential backoff, and encrypted platform tokens using AES-256-GCM.
- Verified inbound delivery webhooks with HMAC-SHA256 signatures and timing-safe comparison, rejecting forged/tampered payloads.
- Built a fragment-based caption composer that generated platform-tailored captions (warm/visual/emoji for Instagram, sharp/280-char for X) from a single blog post input.
- Designed the durable job queue to survive crashes – stuck jobs reset to pending on restart, so no publish silently disappears.

AI-Powered Business Dashboard | *JavaScript, SQL, Dashboard Reporting*

- Designed and built a full-featured ROI calculation and business performance analytics dashboard.
- Implemented financial metrics computation, KPI tracking, and automated reporting logic.
- Delivered actionable insights to support business decision-making through visual data representation.
- Processed and analyzed large datasets to generate performance benchmarks and trend reports.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, Java, C/C++, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Responsive/Accessible UI Design, Figma

Backend/Data: Node.js, Express, Next.js (App Router), REST API Design, MongoDB, SQL

Tools: Git/GitHub, VS Code, Postman, Google Colab

Education

Sardar Vallabhbhai Global University

Ahmedabad, India

B.Tech in Computer Science & Engineering; CGPA: 6.2/10

2024 – Present